BUILD 3

Personal AI Study Kit

Quantitative Aptitude • Logical Reasoning • Verbal Ability

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Course	IMCA — 2nd Year
Bootcamp	AI Readiness 4.0 — Day 3: Learn Smarter with AI
Study topic	Placement Aptitude Preparation

This kit contains study notes, a 10-question quiz with answers, 22 flashcards, source links to check, and the AI prompts used to prepare the material.

1. Study Notes

Quantitative Aptitude tests numerical accuracy, speed, and problem-solving. Common topics include percentages, ratios, averages, time and work, profit and loss, speed-distance-time, and interest.

Logical Reasoning tests pattern recognition, deduction, sequencing, and analytical thinking. Common topics include series, coding-decoding, blood relations, directions, syllogisms, seating arrangements, and puzzles.

Verbal Ability tests grammar, vocabulary, reading comprehension, and sentence correctness. Common topics include error spotting, fill-in-the-blanks, synonyms/antonyms, para jumbles, and sentence correction.

Key formulas

Percentage	(Part ÷ Whole) × 100
Percentage change	((New – Old) ÷ Old) × 100
Successive change	a + b + (ab ÷ 100), using signed values for a and b
Profit / loss	Profit = SP – CP; Loss = CP – SP; percentage = (profit or loss ÷ CP) × 100
Discount	SP = MP × (100 – Discount %) ÷ 100
Average	Sum of values ÷ Number of values
Ratio	For A:B = 2:3, let A = 2x and B = 3x
Time and work	One-day work = 1/a; two workers' combined time = ab/(a+b)
Speed	Distance = Speed × Time; Speed = Distance ÷ Time
Simple interest	SI = (P × R × T) ÷ 100
Compound amount	A = P(1 + R/100)^T for annual compounding

Problem-solving checklist

• Identify what is given: CP/SP, rate, time, work, distance, or ratio.

• Convert all quantities to the same unit.

• Useful fractions: 25% = 1/4, 20% = 1/5, 12.5% = 1/8.

• For work problems, consider taking total work as the LCM of completion times.

• Check whether the result is reasonable.

Logical reasoning: quick frameworks

• **Series:** check differences, ratios, alternating terms, squares/cubes, and alphabet positions.

• **Coding-decoding:** compare alphabet positions, shifts, reversals, and substitutions.

• **Blood relations:** draw a family tree and label relationships.

• **Directions:** sketch a compass: North up, East right, South down, West left.

• **Syllogisms:** use Venn diagrams; conclusions must hold in every valid diagram.

• **Seating:** fix a confirmed position, then apply relative clues.

• **Statement-conclusion:** rely only on what the statement supports.

Verbal ability: quick rules

• **Subject–verb agreement:** the verb must agree with the subject in number.

• **Tense:** keep tense consistent unless meaning requires a change.

• **Prepositions:** learn combinations such as good at, interested in, capable of, depend on.

• **Reading comprehension:** locate keywords and choose an option supported by the passage.

• **Para jumbles:** identify the opening sentence, pronoun references, and cause-effect links.

2. Quiz — 10 MCQs

Choose your answers before checking the key.

1. A shirt costs ■800. Its price increases by 25%, then decreases by 20%. Final price?

A) ■760 B) ■800 C) ■820 D) ■840

2. A finishes a task in 12 days and B in 18 days. Working together, how long?

A) 6.8 days B) 7.2 days C) 7.5 days D) 8 days

3. A train travels 240 km in 3 hours. Average speed?

A) 60 km/h B) 70 km/h C) 75 km/h D) 80 km/h

4. If A:B = 3:4 and B:C = 2:5, what is A:B:C?

A) 3:4:5 B) 3:4:10 C) 6:8:20 D) 3:4:8

5. An item sells for ■960 at 20% profit. Cost price?

A) ■768 B) ■800 C) ■820 D) ■850

6. If CAT is coded as DBU, how is DOG coded?

A) EPH B) EPI C) EPH D) FQI

7. Riya says: “He is the son of my grandfather’s only son.” How is he related to Riya?

A) Father B) Brother C) Uncle D) Cousin

8. A person walks 5 km north, 3 km east, then 5 km south. Distance from start?

A) 3 km B) 5 km C) 8 km D) 13 km

9. Next number: 2, 6, 12, 20, 30, ?

A) 36 B) 40 C) 42 D) 44

10. Choose the correctly spelled word:

A) Accomodate B) Acommodate C) Accommodate D) Acomodate

Answer key and explanations

1. B — ■800. 800 × 1.25 = 1,000; 1,000 × 0.80 = 800.

2. B — 7.2 days. 1/12 + 1/18 = 5/36 work per day; total time = 36/5 = 7.2 days.

3. D — 80 km/h. Speed = 240 ÷ 3 = 80 km/h.

4. C — 6:8:20. Make B equal: 3:4 becomes 6:8; 2:5 becomes 8:20.

5. B — ■800. ■960 is 120% of cost price; CP = 960 × 100 ÷ 120 = ■800.

6. A or C — EPH. Each letter moves forward by one: D→E, O→P, G→H. Options A and C are duplicated in the supplied question.

7. B — Brother, assuming grandfather’s only son is Riya’s father. His son would be Riya’s brother.

8. A — 3 km. North and south movements cancel; 3 km east remains.

9. C — 42. Differences are +4, +6, +8, +10; next is +12.

10. C — Accommodate. Correct spelling uses double c and double m.

3. Flashcards — 22 Cards

Cover the answer and recall it from memory.

1. Percentage formula	(Part ÷ Whole) × 100
2. Percentage increase	((New – Old) ÷ Old) × 100
3. Successive percentage change	a + b + (ab ÷ 100), using signed values
4. Profit percentage	((SP – CP) ÷ CP) × 100
5. Loss percentage	((CP – SP) ÷ CP) × 100
6. Selling price after discount	MP × (100 – discount %) ÷ 100
7. Average formula	Sum of values ÷ number of values
8. Two workers' combined time	ab ÷ (a + b), for times a and b
9. Work-rate shortcut	If A takes a days, one-day work = 1/a
10. Distance formula	Distance = Speed × Time
11. Speed formula	Speed = Distance ÷ Time
12. Simple interest	(P × R × T) ÷ 100
13. Compound amount	P(1 + R/100)^T for annual compounding
14. Ratio method	For A:B = m:n, let A = mx and B = nx
15. Number-series first check	Find differences between consecutive terms
16. Coding-decoding first check	Compare alphabet positions and letter shifts
17. Blood-relation method	Draw a family tree before answering
18. Direction-sense method	Compass: N up, E right, S down, W left
19. Syllogism method	Use Venn diagrams; conclusion must hold in every valid case
20. Seating strategy	Fix one definite position, then apply relative clues
21. Subject–verb agreement	Verb must agree with the subject in number
22. Reading-comprehension strategy	Locate keywords and choose what the passage supports

4. Verified Sources

These sources came from the Perplexity result you provided. To meet the assignment requirement, open each link yourself, confirm it loads and matches the description, and record the date checked. This document does not falsely claim that those pages were independently opened during preparation.

Resource / URL	What it offers	Why useful	Checked
IndiaBIX — Aptitude https://www.indiabix.com/aptitude/questions-and-answers/	Aptitude, logical reasoning, verbal ability, and explanations.	Good for daily topic-wise practice and timed quizzes.	■ Opened Date: _____
GeeksforGeeks — Aptitude for Placements https://www.geeksforgeeks.org/aptitude/aptitude-for-placements/	Placement guides for quantitative aptitude, reasoning, verbal ability, and mock tests.	Useful for structured placement preparation.	■ Opened Date: _____
GeeksforGeeks — Aptitude Questions and Answers https://www.geeksforgeeks.org/aptitude/aptitude-questions-and-answers/	Topic-wise solved questions and quizzes.	Helpful for step-by-step methods and formula revision.	■ Opened Date: _____
Testbook — Mock Tests https://testbook.com/test-series	Mock tests and exam-style practice; select relevant aptitude tests.	Useful for timed practice and performance analysis.	■ Opened Date: _____
IndiaBIX — Verbal Ability Tests https://www.indiabix.com/verbal-ability/questions-and-answers/	Grammar, comprehension, vocabulary, sentence correction.	Useful for verbal ability practice.	■ Opened Date: _____

Source verification steps

■ Open each URL in your browser.

■ Confirm the page loads and the content matches the description.

■ Check formula details against a source page; record the date checked.

■ If a URL redirects or no longer works, replace it with a current relevant page and update the table.

5. Prompts Used

The prompts below are structured examples based on the material you shared. Edit them to match the exact prompts you actually entered in Perplexity before submitting.

Prompt 1 — Generate study notes

Role: You are a university-level placement aptitude tutor.
Task: Create study notes on quantitative aptitude, logical reasoning, and verbal ability.
Context: I am an IMCA second-year student preparing for internships and placement tests.
Constraints: Use beginner-friendly English, include essential formulas and practical methods, and avoid unsupported claims.
Format: Use headings, bullet points, and a compact formula table.

Prompt 2 — Create quiz

Role: You are an aptitude-test question setter.
Task: Create 10 MCQs covering quantitative aptitude, logical reasoning, and verbal ability.
Context: I need beginner-to-intermediate placement practice.
Constraints: Give four options per question and one correct answer; include calculations for numerical questions.
Format: Numbered quiz followed by an answer key with short explanations.

Prompt 3 — Create flashcards

Role: You are a study-card designer.
Task: Turn the key concepts and formulas from the notes into at least 20 flashcards.
Context: I want to revise efficiently before aptitude tests.
Constraints: Each card should test one idea, keep answers concise, and preserve formula accuracy.
Format: Numbered Front/Back cards.

Prompt 4 — Find sources

Role: You are a research assistant.
Task: Find reliable sources for placement aptitude, logical reasoning, and verbal ability.
Context: The sources support a student study kit.
Constraints: Prefer established education websites, give direct URLs, and do not call sources verified unless opened and checked.
Format: Table with resource, URL, content, and usefulness.

Prompt 5 — Verify answers

Role: You are an accuracy-focused aptitude tutor.
Task: Check quiz answers and explanations for calculation or wording errors.
Context: The quiz is for an IMCA placement revision kit.
Constraints: Show key calculations and flag ambiguity or duplicate options.
Format: Numbered list of answer, verification, and correction if needed.

Final submission checklist

■ Study notes with headings included.

■ At least 10 quiz questions with answers included.

■ At least 20 flashcards included (22 in this kit).

■ Open and check each source; fill the source-verification dates honestly.

■ Edit prompts to match the actual prompts used in Perplexity.

■ Submit using the form in the AI Readiness 4.0 WhatsApp group by the stated deadline.